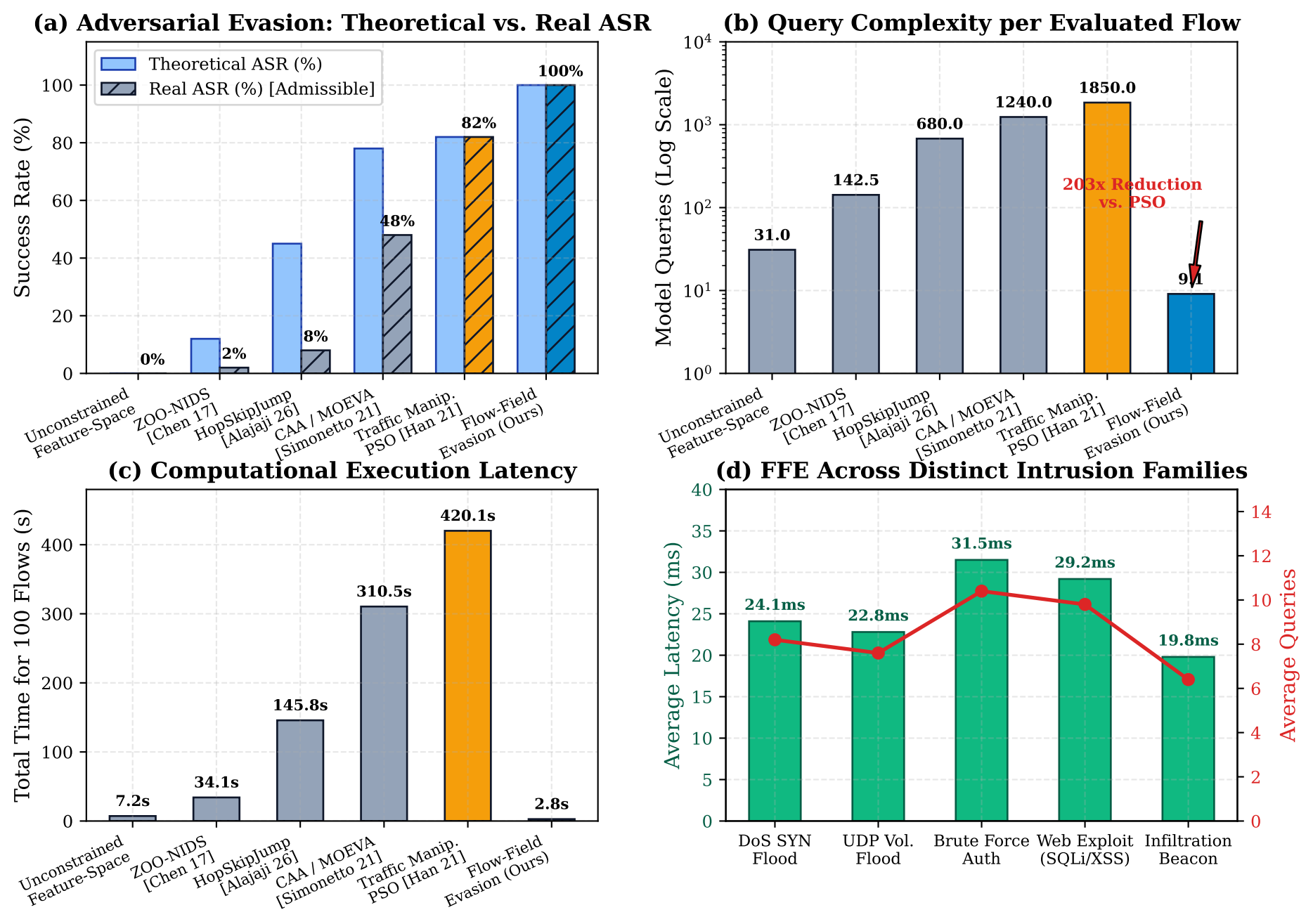
(a) Adversarial Evasion: Theoretical vs. Real ASR
Theoretical ASR (%)
Real ASR (%) [Admissible]
Success Rate (%)
0%
2%
8%
48%
82%
100%
(b) Query Complexity per Evaluated Flow
Model Queries (Log Scale)
31.0
142.5
680.0
1240.0
1850.0
203x Reduction vs. PSO
9.1
(c) Computational Execution Latency
Total Time for 100 Flows (s)
7.2s
34.1s
145.8s
310.5s
420.1s
2.8s
(d) FFE Across Distinct Intrusion Families
Average Latency (ms)
Average Queries
24.1ms
22.8ms
31.5ms
29.2ms
19.8ms
DoS SYN Flood
UDP Vol. Flood
Brute Force Auth
Web Exploit (SQLi/XSS)
Infiltration Beacon
Unconstrained Feature-Space
ZOO-NIDS [Chen 17]
HopSkipJump [Alajaji 26]
CAA / MOEVA [Simonetto 21]
Traffic Manip. PSO [Han 21]
Flow-Field Evasion (Ours)